Stationary Distribution and Mixing of a Positive Finite Markov Chain

An End-to-End Meta-Math Walkthrough

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(Dated: October 2026)

Abstract

Every row contains the same positive uniform component. Removing it leaves a stochastic remainder that cannot enlarge distance, so each step contracts by a common factor. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PS-05 / Probability and Stochastic Processes / Report v1

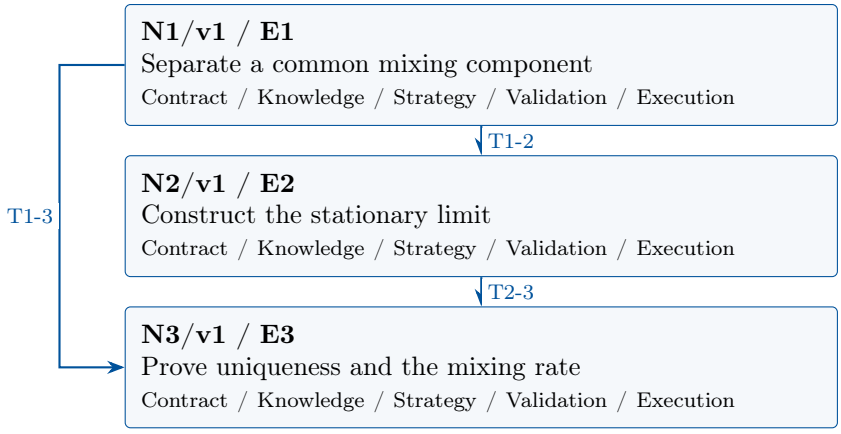
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $P = (P_{ij})_{i,j=1}^{d}$ be a row-stochastic matrix with every entry strictly positive. For row probability vectors, prove that there is a unique stationary distribution π with $\pi P = \pi$, and $\|\mu P^n - \pi\|_1 \le (1-\delta)^n \|\mu - \pi\|_1$ for every initial distribution μ and $n \ge 0$, where $\varepsilon = \min_{i,j} P_{ij}$ and $\delta = d\varepsilon \in (0, 1]$. If $\delta = 1$, convergence is exact after one step; the exponent-zero factor at $n = 0$ is interpreted as one.*

A. Foundations and permitted imports

Use completeness of finite-dimensional Euclidean space, closedness of the probability simplex, the triangle inequality, and the geometric series. Total variation distance on this finite set is one half of the ℓ^1 distance. No spectral or Perron–Frobenius theorem is imported; the fixed point is constructed by a Cauchy estimate.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. If $\delta < 1$, $P = \delta U + (1-\delta)Q$ for a stochastic Q, and $\|\mu P - \nu P\|_1 \leq (1-\delta)\|\mu - \nu\|_1$.

Receiving activity. N2.1

Compatibility. E1/v1 with $q = 1 - \delta \in [0, 1)$ and $\mu_n = \mu_0 P^n$. Check: Consecutive differences tending to zero is not enough; the summable geometric tail provides the Cauchy property.

T1-3: N1/E1/v1 to N3. If $\delta < 1$, $P = \delta U + (1-\delta)Q$ for a stochastic Q, and $\|\mu P - \nu P\|_1 \leq (1-\delta)\|\mu - \nu\|_1$.

Receiving activity. N3.1

Compatibility. E1/v1 and a stationary π from E2/v1. Check: The norm is on row distributions, consistent with the stochastic convention; no column convention is silently substituted.

T2-3: N2/E2/v1 to N3. μ_n converges to a probability vector π with $\pi P = \pi$.

Receiving activity. N3.1

Compatibility. E1/v1 and a stationary π from E2/v1. Check: The norm is on row distributions, consistent with the stochastic convention; no column convention is silently substituted.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Separate a common mixing component

Object and role. Write P as a convex combination of a uniform reset and a stochastic residual.
Local obligation: If $\delta < 1$, $P = \delta U + (1-\delta)Q$ for a stochastic Q, and $\|\mu P - \nu P\|_1 \le (1-\delta)\|\mu - \nu\|_1$.

Inputs. $\varepsilon > 0$, $\delta = d\varepsilon$, and the matrix U whose every entry is $1/d$.

C — Contract

Publish E1/v1 only after establishing this obligation: If $\delta < 1$, $P = \delta U + (1-\delta)Q$ for a stochastic Q, and $\|\mu P - \nu P\|_1 \le (1-\delta)\|\mu - \nu\|_1$.

K — Knowledge

$\varepsilon > 0$, $\delta = d\varepsilon$, and the matrix U whose every entry is $1/d$. Retain the stated assumptions and import versions.

S — Strategy

Write P as a convex combination of a uniform reset and a stochastic residual.

V — Validation

The case $\delta = 1$ must be handled before dividing by $1 - \delta$.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Row sums give $d\varepsilon \le 1$. If $\delta = 1$, every row entry must equal $1/d$, so $P = U$. Otherwise define $Q_{ij} = (P_{ij} - \varepsilon)/(1-\delta)$. Its entries are nonnegative and each row sums to one, proving the decomposition.

N1.2 For any row vector z, $\|zQ\|_1 \le \sum_j \sum_i |z_i| Q_{ij} = \sum_i |z_i| = \|z\|_1$. For probability vectors μ, ν, $(\mu - \nu)U = 0$, since their coordinates both sum to one. The decomposition therefore gives the contraction.

Export E1/v1. If $\delta < 1$, $P = \delta U + (1-\delta)Q$ for a stochastic Q, and $\|\mu P - \nu P\|_1 \le (1-\delta)\|\mu - \nu\|_1$.

Acceptance gate. The case $\delta = 1$ must be handled before dividing by $1 - \delta$.

Targeted repair. If strict positivity is replaced by nonnegativity, check whether ε becomes zero; the strict contraction may then disappear. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Construct the stationary limit

Object and role. Bound consecutive differences geometrically and sum the tail to prove convergence inside the simplex. **Local obligation:** μ_n converges to a probability vector π with $\pi P = \pi$.

Inputs. E1/v1 with $q = 1 - \delta \in [0, 1)$ and $\mu_n = \mu_0 P^n$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: μ_n converges to a probability vector π with $\pi P = \pi$.

K — Knowledge

E1/v1 with $q = 1 - \delta \in [0, 1)$ and $\mu_n = \mu_0 P^n$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Bound consecutive differences geometrically and sum the tail to prove convergence inside the simplex.

V — Validation

Consecutive differences tending to zero is not enough; the summable geometric tail provides the Cauchy property.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 For $q \in (0, 1)$, contraction gives $\|\mu_{n+1} - \mu_n\|_1 \leq q^n \|\mu_1 - \mu_0\|_1$. Therefore for $m > n$, $\|\mu_m - \mu_n\|_1 \leq q^n \|\mu_1 - \mu_0\|_1/(1 - q)$. This tends to zero uniformly in $m > n$, so the sequence is Cauchy.

N2.2 The simplex is closed in a complete finite-dimensional space, so the limit π is a probability vector. Multiplication by P is continuous; taking limits of $\mu_{n+1} = \mu_n P$ yields stationarity. If $q = 0$, $P = U$ and every distribution becomes the uniform distribution in one step, giving the same conclusion directly.

Export E2/v1. μ_n converges to a probability vector π with $\pi P = \pi$.

Acceptance gate. Consecutive differences tending to zero is not enough; the summable geometric tail provides the Cauchy property.

Targeted repair. If convergence is inferred from small single increments, insert the displayed tail sum; it is the missing uniform bound over later indices. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Prove uniqueness and the mixing rate

Object and role. Iterate the contraction against a fixed point and compare any two fixed points. **Local obligation:** π is unique and the stated geometric mixing estimate holds for all initial distributions.

Inputs. E1/v1 and a stationary π from E2/v1. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: π is unique and the stated geometric mixing estimate holds for all initial distributions.

K — Knowledge

E1/v1 and a stationary π from E2/v1. Incoming proof exports: E1/v1 at N3.1, E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Iterate the contraction against a fixed point and compare any two fixed points.

V — Validation

The norm is on row distributions, consistent with the stochastic convention; no column convention is silently substituted.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 If π' is another stationary distribution, contraction gives $\|\pi - \pi'\|_1 \le q\|\pi - \pi'\|_1$. Since $q < 1$, the norm is zero, proving uniqueness.

N3.2 For any μ, repeatedly compare μP^n with $\pi P^n = \pi$ to obtain $\|\mu P^n - \pi\|_1 \le q^n \|\mu - \pi\|_1$. When $q = 0$ the direct one-step conclusion applies. Thus existence, uniqueness, and the quantitative rate are all established.

Export E3/v1. π is unique and the stated geometric mixing estimate holds for all initial distributions.

Acceptance gate. The norm is on row distributions, consistent with the stochastic convention; no column convention is silently substituted.

Targeted repair. If existence of a stationary vector is assumed while proving convergence, use E2's actual limit construction before invoking stationarity in the estimate. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Separate a common mixing component

1. **N1.1.** Row sums give $d\varepsilon \le 1$. If $\delta = 1$, every row entry must equal $1/d$, so $P = U$. Otherwise define $Q_{ij} = (P_{ij} - \varepsilon)/(1 - \delta)$. Its entries are nonnegative and each row sums to one, proving the decomposition.

2. **N1.2.** For any row vector z, $\|zQ\|_1 \le \sum_j \sum_i |z_i| Q_{ij} = \sum_i |z_i| = \|z\|_1$. For probability vectors μ, ν, $(\mu - \nu)U = 0$, since their coordinates both sum to one. The decomposition therefore gives the contraction.

B. N2 — Construct the stationary limit

1. **N2.1.** For $q \in (0, 1)$, contraction gives $\|\mu_{n+1} - \mu_n\|_1 \le q^n \|\mu_1 - \mu_0\|_1$. Therefore for $m > n$, $\|\mu_m - \mu_n\|_1 \le q^n \|\mu_1 - \mu_0\|_1/(1 - q)$. This tends to zero uniformly in $m > n$, so the sequence is Cauchy.

2. **N2.2.** The simplex is closed in a complete finite-dimensional space, so the limit π is a probability vector. Multiplication by P is continuous; taking limits of $\mu_{n+1} = \mu_n P$ yields stationarity. If $q = 0$, $P = U$ and every distribution becomes the uniform distribution in one step, giving the same conclusion directly.

C. N3 — Prove uniqueness and the mixing rate

1. **N3.1.** If π' is another stationary distribution, contraction gives $\|\pi - \pi'\|_1 \le q\|\pi - \pi'\|_1$. Since $q < 1$, the norm is zero, proving uniqueness.

2. **N3.2.** For any μ, repeatedly compare μP^n with $\pi P^n = \pi$ to obtain $\|\mu P^n - \pi\|_1 \le q^n \|\mu - \pi\|_1$. When $q = 0$ the direct one-step conclusion applies. Thus existence, uniqueness, and the quantitative rate are all established.

□

V. WORKED ORCHESTRATION

For $P = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{smallmatrix}\right)$, $\varepsilon = 1/4$, $\delta = 1/2$, and $q = 1/2$. Solving $\pi P = \pi$ gives $\pi = (2/3, 1/3)$. For $\mu = (1, 0)$, the initial ℓ^1 distance is $2/3$; after one step it is $|3/4 - 2/3| + |1/4 - 1/3| = 1/6 \le (1/2)(2/3)$.

A. Boundary and hypothesis audit

The nonnegative stochastic matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ has stationary distribution $(1/2, 1/2)$ but alternates forever from $(1, 0)$. Strict positivity is a sufficient mixing assumption here. It is not necessary for all mixing chains, but dropping it without a substitute invalidates this argument.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 claim: every finite irreducible Markov chain converges to stationarity. The two-state alternating chain is irreducible but periodic. **Diagnosis:** Knowledge omitted the mixing condition. **Repair:** use the strict-positive-entry contract, which yields $q < 1$ directly. Recheck the geometric tail and the final rate; a broader irreducible aperiodic theorem would need a separate proof.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Extend the same method when a power P^m is strictly positive, yielding mixing bounds in blocks of m steps.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Amir Dembo, Probability Theory: STAT310/MATH230 notes for measure-theoretic probability and limit arguments; Stanford STAT310C syllabus and lecture materials for martingale inequalities and stopping theorems. The versions and proofs used in this collection are stated explicitly.